Step 3: Analyze — Method Record

CAPSTONE-01 | AI-Assisted Audit Control Framework · MCX AI Lab: Data Analysis with AI
Dataset: **grc_risk_register.csv** — synthetic GRC risk register, 160 rows × 19 columns · snapshot date 30 Jun 2026

1 PURPOSE AND DELIVERABLE

This record documents **how** the Step 3 analysis was produced — what was run, what was decided, what was corrected, and what remains unproven. It is the audit trail behind the artifact, not the artifact itself. Findings and exhibits live in **CAPSTONE-01_Risk_Register_Dashboard.html** and the five-page board pack **CAPSTONE-01_Step3_Dashboard_Exhibit_Pack.pdf**, both generated from a single computation run so they cannot drift from each other or from the data.

2 WHAT WAS RUN

Populations	Descriptive measures on all **160** items; all timeliness measures on the **116** items due on or before the snapshot, with the **44** not-yet-due items excluded from every timeliness denominator per the Step 2 rule. Breach rate carried on both bases: 20.6% of the register, 28.4% of items that faced a deadline.
Comparisons	Cross-sectional only. Six segment fields — department, domain, owner role, tier, identification source, control effectiveness — each against the eligible-population mean, plus a closed-versus-breached contrast. No period-over-period claims, no external benchmark.
Severity metric	Overrun ratio = days past due ÷ original deadline window (due – identification date), reported with absolute days. Tier-independent by design, so it survives if the tier artifact is confirmed.
Money	Exposure summarised by state, aging band, control rating and tier. Treated throughout as **magnitude** — how much sits where — never as an explanatory variable.
Method	pandas over the source CSV; every figure in both artifacts traces to that run, none transcribed by hand.

3 WHAT THE ANALYSIS FOUND

CRITICAL	**Control effectiveness is the strongest discriminator in the file.** Breach runs 5.6% (Effective) to 80.0% (Ineffective) — a fourteen-fold spread, independent of risk tier. Of the $42.1M past due, **$41.6M (98.8%) sits behind a control not rated Effective**.
CRITICAL	**“Not Tested” is not behaving as a neutral state.** Untested controls breach at 42.1%, indistinguishable from Partially Effective at 42.4%. Compounded by the Step 1 finding that **17 items closed while their control was still untested**.
MATERIAL	**Aged exposure is heavily concentrated.** 21 of 33 past-due items are beyond 180 days and carry $39.8M — 94.6% of past-due exposure. Recent lapses are numerous but small; the money is in the long tail.
MATERIAL	**Regulator-identified items are the least likely to close on time.** External Regulatory Exam breaches at 46.7% against 11.8% for Third-Party Review. Base of 15 is thin and reported as such, but this is the most exposed line in the pack for an audit committee.
WATCH	**Severity inversion survives the denominator correction.** High 36.6% and Critical 33.3% against Low 20.0%; median overruns 2.20× and 2.94× against 0.27×. Artifact risk unresolved, carried to Verify.
WATCH	**Exposure concentration flagged, not featured.** Past-due items are 21% of count but 29% of dollars. Exposure is banded from tier, so this restates tier concentration rather than adding evidence; the caveat is attached in three places.

4 DECISIONS AND CORRECTIONS MADE DURING THE BUILD

Scope extended	Exposure was added to every exhibit at reviewer direction after the initial findings run — accepted with a standing guardrail, not as a free addition. Exposure is tier-banded, so it is shown as magnitude per state and never as a driver, with the caveat repeated in three places.
Self-correction	The population had been characterised as bifurcated with “nothing in between.” Overstated, and withdrawn. Actual eligible-population distribution: 65 on time or early, 25 at 1–90 days, 5 at 90–180, 21 beyond 180 — and items closed late (to 59 days) overlap the newest breaches (from 13 days). Restated as **two populations with a thin transition zone**.
Provenance challenge	The reviewer challenged the source of the “Not Tested” category. Traced to **column 10, Control_Effectiveness** — a four-value categorical with zero nulls, neither derived nor inferred. The exhibit shows a base of 19 rather than the file total of 27 because the denominator rule removes 8 items not yet due.
Anomaly surfaced	That check exposed a control-rule gap: items carry “Not Tested” alongside Status “Closed – Validated” (e.g. GRC-2026-0035, GRC-2025-0036). A register with enforced control rules would normally make that combination impossible. Added to the Verify agenda.
Rendering	The board pack was built natively in reportlab, not converted from HTML — the local converter has a known flex and grid limitation that collapses the bar tracks. A page-numbering defect (hardcoded total of four against an actual five) was caught in review and replaced with a two-pass count.
Presentation	Each exhibit header carries its denominator as a literal fraction. This puts the central Step 2 decision on the face of the document rather than in a footnote — it is the most likely point of challenge in the room.

5 WHAT THIS STEP DOES NOT ESTABLISH

Not trend	Status is assigned independently of how long an item has existed, making period-over-period comparison invalid. No improvement or deterioration claim is made anywhere in the pack.
Not tolerance	No internal threshold or external benchmark exists in this data and none has been assumed. The analysis does not state whether a 20.6% breach rate is acceptable.
Not causation	Control rating is shown to be **associated** with on-time closure. Whether it drives that outcome, and whether the relationship is genuine or a generator artifact, is untested.
Not remediation	The pack describes the position. Setting tolerances, prioritising items and directing remediation sit with senior leadership once findings are presented.

6 CARRY-FORWARD TO STEP 4 (VERIFY)

Four adversarial tests, in priority order. **(1) Tier integrity** — tier is a direct banding of the inherent score and exposure is banded the same way; test whether the tier-to-overdue pattern reflects behaviour or construction. **(2) Control-rating challenge** — the fourteen-fold spread is strong enough to be suspicious; test whether the generator wrote it in. **(3) Residual haircut** — the Step 1 hypothesis that residual is a mechanical reduction off inherent; group by control effectiveness and compare reduction ratios. This is a **separate claim** from test 2: a rating can predict closure timing while having nothing to do with how residual was computed. **(4) Control-rule integrity** — whether “Not Tested” can legitimately coexist with “Closed – Validated.”

METHOD AND CONFIDENCE. Confidence in the structural findings is high — arithmetic over a clean file with stated denominators. Confidence that the observed relationships reflect real behaviour rather than generator design is **deliberately withheld** pending Step 4. Bases under 20 are marked thin and are not treated as conclusions about any named unit.
SOURCE AND ACCOUNTABILITY. Synthetic GRC risk register for portfolio demonstration; no employer data. Analysis and drafting performed with Claude (Opus 5) under human direction; every figure regenerated from source, none transcribed. Reviewed, verified and owned by N. Fragoso.